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Research Paper

A knowledge graph for standard carbonate microfacies and its application in the automatical reconstruction of the relative sea-level curve

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ABSTRACT

The reconstruction of high-resolution sea-level variation curves in deep time based on the standard carbonate microfacies knowledge graph (SMFKG) is of great scientific significance for exploring the Earth system evolution and predicting future sea-level and climate changes. In this study, the concepts, attributes, and relationships among standard carbonate microfacies (SMF) are comprehensively analyzed; an ontology layer is established and its data layer is constructed using thin-section descriptions; and finally, the SMFKG is established. Additionally, based on the knowledge graph, an application for automatically identifying SMF using identification markers and reconstructing the high-resolution relative sea-level variation curve using the SMF and facies zones is compiled. Then, all thin sections of the late Ediacaran Dengying Formation in the western margin of the Yangtze Platform are observed and described in detail, the SMF and facies zones are identified automatically, and the relative sea-level curve is reconstructed automatically using the SMFKG. The reconstruction results show that the Yangtze Platform experienced four sea-level rise and fall cycles in the late Ediacaran, of which two intense regressions led to subaerial-exposed unconformities in the interior and top of the Dengying Formation, which is highly consistent with previous research results. This shows that the high-resolution relative sea-level variation curve in deep time can be reconstructed efficiently and intelligently using the SMFKG. Additionally, in the near future, the combination of an automatic digital slide-scanning system, machine-learning techniques, and the SMFKG can achieve one-stop fully automatic SMF recognition and reconstruction of high-resolution relative sea-level variation curves in deep time, which has a high application value.

1. Introduction

Sea-level changes are one of the most important indicators of global climate and environmental change, which have an important impact on normal human life and socioeconomic activities (Cazenave and Nerem, 2004; Chen et al., 2013; Gardner et al., 2013). Eustatic sea-level variation in deep time have important scientific value for discussing the coordinated evolutions of paleostructure, paleogeography, paleoocean, paleontology and paleoclimate, as well as the distribution law of minerals and predicting sea-level and climate trends (Haq et al., 1987; Haq and Schutter, 2008; Geyman et al., 2021). Because relative sea-level variations are the comprehensive result of tectonic activity, eustatic sea-level changes, and sediment supply, the analysis of relative sea-level variation in sedimentary areas with stable tectonic settings, such as a rimmed carbonate platform or a homoclinal carbonate ramp, can indirectly reflect the eustatic sea-level variations in deep time (Watts, 1982; Villaseñor et al., 2015; Mestdagh et al., 2019). Carbonate microfacies and their combinations can indicate the sedimentary environment and high-resolution carbonate microfacies analysis can recreate sedimentary

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environment variations, which is an important method for reconstructing relative sea-level variation curves in deep time (Hallam, 2001; Flügel, 2004, 2010; Carević et al., 2013; Xi et al., 2016). Through the observation and quantitative statistical analyses of lithology, texture, sedimentary structure, and fossil information in thin sections, the carbonate microfacies can be identified and, then, the relative sea-level variations in deep time can be obtained (Han et al., 2016; Li et al., 2017; Zhang et al., 2018). However, there still exist some obstacles to the wide application of carbonate microfacies for reconstructing high-resolution relative sea-level variation curves in deep time. These include ununified types of carbonate microfacies; lack of semi-quantitative or quantitative constraints on various identification markers (IDM) for carbonate microfacies, which are affected by subjective judgments; and a large quantity of data is required to identify all carbonate microfacies for outcrops with numerous thin sections, and the process of reconstructing high-resolution sea-level curves is cumbersome.

A knowledge graph is a new research method used for managing large quantities of knowledge and other intelligent services in the big data era (Tiwari et al., 2021). In the process of processing massive data quantities, a knowledge graph connects the upstream work of knowledge engineering and representation, the midstream work of data integration, and the downstream work of data analysis and result communication (Wing, 2019; Ma, 2021). This method is good at revealing the source and development law of knowledge in complex knowledge fields through data mining, information processing, knowledge metrology and graphing (Jordan and Mitchell, 2015; Ji et al., 2021). Particularly, the powerful data processing and logical reasoning functions of a knowledge graph provide ideal technical means to address the obstacles in the carbonate microfacies reconstruction of relative sea-level variation curves in deep time.

In this study, the ontology and data layers of standard carbonate microfacies (SMF) are constructed and used to establish a knowledge graph. Based on the standard carbonate microfacies knowledge graph (SMFKG), the automatic identification of SMF using the thin-section descriptions, discrimination of facies zones (FZ) based on SMF combinations, and reconstruction of relative sea-level curves in deep time from those SMF and FZ is compiled. Then, the SMFKG is used to automatically identify the SMF of all thin sections from the Dengying Formation at the western margin of the Yangtze Platform and automatically reconstruct its high-resolution relative sea-level curve at the end of the Ediacaran. Additionally, the application prospects of the SMFKG are also discussed.

2. Background knowledge

Different types of carbonate sedimentary models have been developed in different tectonic settings for the long geological history of Earth's evolution. These different carbonate sedimentary models had different carbonate microfacies assemblages. Through the comprehensive analysis of carbonate sedimentary models and SMF, their core elements (including rimmed carbonate platforms and homoclinal carbonate ramps, see Section 2.1) and standard types (including the SMF of such settings, see Section 2.2) are determined; a rimmed carbonate platform is the focus of this paper.

2.1. The carbonate sedimentary models

Predecessors have carried out a lot of research work on carbonate sedimentary models. Before the 1950s, most scholars believed that most carbonate sedimentary environments were shallow water and that carbonate rocks were the product of chemical sedimentation. However, with the gradual understanding and deepening of modern carbonate sedimentary principles from the early 1960s onward, it has been determined that carbonate rocks are products of biological and biochemical processes and their distributions are controlled by hydrological conditions, similar to clastic rocks. On this basis, a series of new explanations have been made for ancient carbonate sedimentary environments, and sedimentary models corresponding to modern carbonate sedimentary environments have been developed. Shaw (1964) first divided the main sedimentary environment of carbonate (shallow sea) into two different types; the epicontinental sea and pericontinental sea. Based on Shaw's epicontinental sea model and the energy distribution of tides and waves, Irwin (1965) identified three energy zones: The X zone far from the coast (low-energy zone), the Y zone slightly nearer the coast (high-energy particle beach zone), and the Z zone near the coast (low-energy zone). Laporte (1969) put more emphasis on the influence of frequent tidal changes in the carbonate sedimentary model and then identified four FZ on the basis of Shaw and Irwin. These were the supratidal and intertidal, intertidal, subtidal (without terrigenous debris), and subtidal (with terrigenous debris) zones. Armstrong (1974) proposed two sedimentary models for clastic rock–carbonate rock and carbonate rock according to different Carboniferous sedimentary assemblages in the Arctic of Alaska, North America; the latter is a comprehensive carbonate platform sedimentary model including nine FZ. Wilson (1975) established a carbonate sedimentary model from the shallow to deep sea environments, which is divided into there major FZ, nine secondary FZ, and twenty-four SMF. This model was not only pioneering at that time but also continues to be used widely today. Tucker's (1981) model includes seven main sedimentary environments and defines a shallow-water carbonate sedimentary system dominated by the platform epicontinental sea and a pelagic carbonate sedimentary system dominated by the slope basin. Based on different tectonic paleogeographic backgrounds, Flügel (2004) preliminarily divided carbonate platforms into eight main facies models: rimmed carbonate shelf, nonrimmed carbonate shelf, homoclinal ramp, distal steepening gentle ramp, isolated platform, marine atoll, epicontinental platform, and underwater platform. Subsequently, some scholars established different carbonate sedimentary models according to comprehensive considerations of carbonate platform evolution, environmental characteristics, microfacies analysis, and other factors (Kanzaki et al., 2021; Xiao et al., 2021).

Controlled by sedimentation, sea-level rise–fall cycles, and tectonics, carbonate rock can form under different sedimentary models, thus it is crucial to establish the classification and evolution process of carbonate sedimentary models (Tucker, 1981; Read, 1982, 1985; Carozzi, 1989). Based on an analysis of previous marine carbonate sedimentary models, Read (1989) proposed the homoclinal carbonate ramp and carbonate platform models, with three end element types—homoclinal slope, bordered shelf, and isolated platform or marine atoll. Carozzi (1989) divided shallow-water carbonate rocks into two end element types, carbonate platform and homoclinal carbonate ramp, and then further divided those into 6 major and 25 secondary sedimentary models according to sedimentary environment differences. It can be seen that Read's and Carozzi's classifications of carbonate sedimentary models emphasized homoclinal carbonate ramp and carbonate platform end elements. Subsequently, geologists gradually eliminated the limitations of static carbonate sedimentary models and began establishing the dynamic evolution process between different carbonate sedimentary models composed of two basic end elements, rimmed carbonate platform and homoclinal carbonate ramp (Read, 1982, 1985; Read and Grotzinger, 1986; Carozzi, 1989).

2.2. The carbonate microfacies schema

Microfacies analysis forms a fundamental and crucial part of the study of sedimentary environments in different carbonate sedimentary models. Udden and Waite (1927) described the practical application of limestone structure in thin sections, while Brown (1943) was the first to define the term *microfacies*, which referred to the composition and appearance of thin sections. Subsequently, Cuvillier (1952) redefined the term *microfacies* as the sum of all paleontological and petrological characteristics of sedimentary thin sections. Dunham (1962) studied microfacies in detail based on the relative contents of micrite or sparry grains and their responses to sedimentary environment energy. Wilson (1975) systematically summarized the research methods of carbonate microfacies. Wilson also established the standard facies belt model of a rimmed carbonate platform including 24 standard microfacies, which became the basis for subsequent scholars. Flügel (1978) comprehensively summarized the limestone microfacies analysis methods. Flügel et al. (1989) further revised the concept of microfacies and considered them the sum of all paleontological and sedimentological markers that can be classified from thin sections. Spence and Tucker (1999) studied the rise and fall of sea level and the environmental changes based on the carbonate microfacies results. Flügel (2004, 2010) comprehensively and finely described the two basic carbonate sedimentary models of rimmed carbonate platform and homoclinal carbonate ramp and divided them into 26 and 30 SMF, respectively.

The five main criteria used by Flügel (2004, 2010) to classify the SMF of rimmed carbonate platforms and homoclinal carbonate ramps were as follows: (1) grain types, frequency, and combinations; (2) matrix types; (3) sedimentary fabrics (such as lamina, grading, open-space structures, burrowing, reworking, and redeposition); (4) fossil types and abundances; and (5) sedimentary textures. Rimmed carbonate platforms are divided into 10 FZ and further divided into 26 SMF. The specific characteristics of the rimmed carbonate platforms are shown in Fig. 1. Additionally, Flügel (2004, 2010) divided the homoclinal carbonate ramp sedimentary model into 4 FZ and 30 SMF. In both models, all FZ and SMF experiences gradual deepening from the nearshore zone to the deep-water basin (Fig. 1). For example, in the rimmed carbonate platform (from SMF1 in the basin facies to SMF26 affected by subaerial-exposed dissolution) and in the homoclinal carbonate ramp (from SMF1 in the basin facies to SMF30 in the inner ramp). However, one microfacies can also be distributed in multiple adjacent FZ, indicating that each microfacies only indicates the relative water-depth range. For example, SMF11 is distributed in two different FZ (FZ5 and FZ6) in the rimmed carbonate platform, indicating that it is in the water depth range corresponding to FZ5-FZ6 (Fig. 1). Flügel (2004, 2010) provided a detailed analysis method for thin-section observation, microfacies analysis, and discrimination; FZ identification; and sedimentary environment interpretation and established a unified standard that has practical application value. This standard laid a solid foundation for unifying the concept of carbonate microfacies, establishing the SMFKG and reconstructing the relative sea-level curve in deep time by using the SMFKG. In this work, the SMF of the rimmed carbonate platform are selected as the basis for constructing the SMFKG.

3. Building the standard carbonate microfacies knowledge graph

A knowledge graph is a graphical representation in which nodes represent entities and edges represent the relationships among these entities (Raskin and Pan, 2005; Hogan, 2020). The knowledge graph data organization structure is composed of an ontology layer and a data layer (Chen et al., 2020). The consideration of whether to build the knowledge graph ontology layer first and then the data layer, or the data layer first and then the ontology layer, respectively represent the top-down and bottom-up processes (Ma, 2021). The top-down construction process is mainly used for the subject domain knowledge graphs, while the bottom-up process is mainly used for the general domain knowledge graphs (Ma and Fox, 2013; Hao et al., 2021). Therefore, the SMFKG is constructed from top to bottom. First, the ontology of SMF is constructed, and then the specific data obtained from different data sources is linked to the corresponding ontology layer. In the final step, the knowledge graph is formed by combining the ontology layer and the data layer (Fig. 2).

3.1. Ontology layer construction

The ontology layer, builds a data model including entities, attributes, and the relationships among them through the analysis of various entities of interest in a subject domain (Buccella et al., 2009; Nickel et al., 2015; Aydin and Aydin, 2020). According to the abovementioned application requirements, for constructing the ontology layer of SMF, the four main considerations are as follows: (1) The facies markers required to identify each microfacies; (2) the specific microfacies identified by those facies markers and the water depth represented by each; (3) the FZ composed of different microfacies; (4) the carbonate sedimentary models composed of the different FZ.

Based on the above considerations, the SMF of the rimmed carbonate platform in Flügel (2004, 2010) and other references (Wilson, 1975) are labeled by domain experts. Furthermore, the entities, relationships and attributes are extracted, including 152 entities, 3 types of relationships and 129 attributes.

3.1.1. Collection of entities

The three main entities (concepts) included are as follows:

(1) FZ: due to the differences of paleogeomorphology, each carbonate sedimentary model includes multiple different FZ, of which the ideal-rimmed carbonate platform model includes a total of 10 (from FZ1 to FZ10).
(2) SMF: different sedimentary models and FZ are composed of different SMF combinations, including 26 SMF (SMF1–SMF26) in the ideal rimmed carbonate platform.
(3) Microfacies (identification) markers: the main facies markers of each microfacies include grain type, texture, sedimentary structure, and fossils (Fig. 3).

3.1.2. Collection of relationships

In the SMFKG, "SMF" are the connection centers of various entities. Therefore, the main relationships are the correlations among SMF and different entities or attributes (Fig. 4). The three main relationships are as follows:

(1) The inclusion relationship refers to how many FZ are included in the carbonate sedimentary model, and the deepening relationship refers to the gradual deepening of each FZ from the coast to the deep sea. For example, the rimmed carbonate platform contains 10 FZ (FZ1–FZ10) from the shallow-water zone to the deep-water basin zone.
(2) An identification markers (IDM) relationship means that a SMF needs multiple facies markers or a FZ needs a specific SMF combination to determine or identify. For example, the SMF can be identified by grain, matrix, texture, sedimentary structure, and so on, while the IDM of the marginal facies zone (FZ6) in the rimmed carbonate platform are composed of SMF11, SMF12S, SMF13, and SMF15.
(3) A composite attribute relationship represents an entity that contains multiple attribute features. For example, grain includes type, content, size, sorting, and rounding; while fossil characteristics include biological type and abundance.

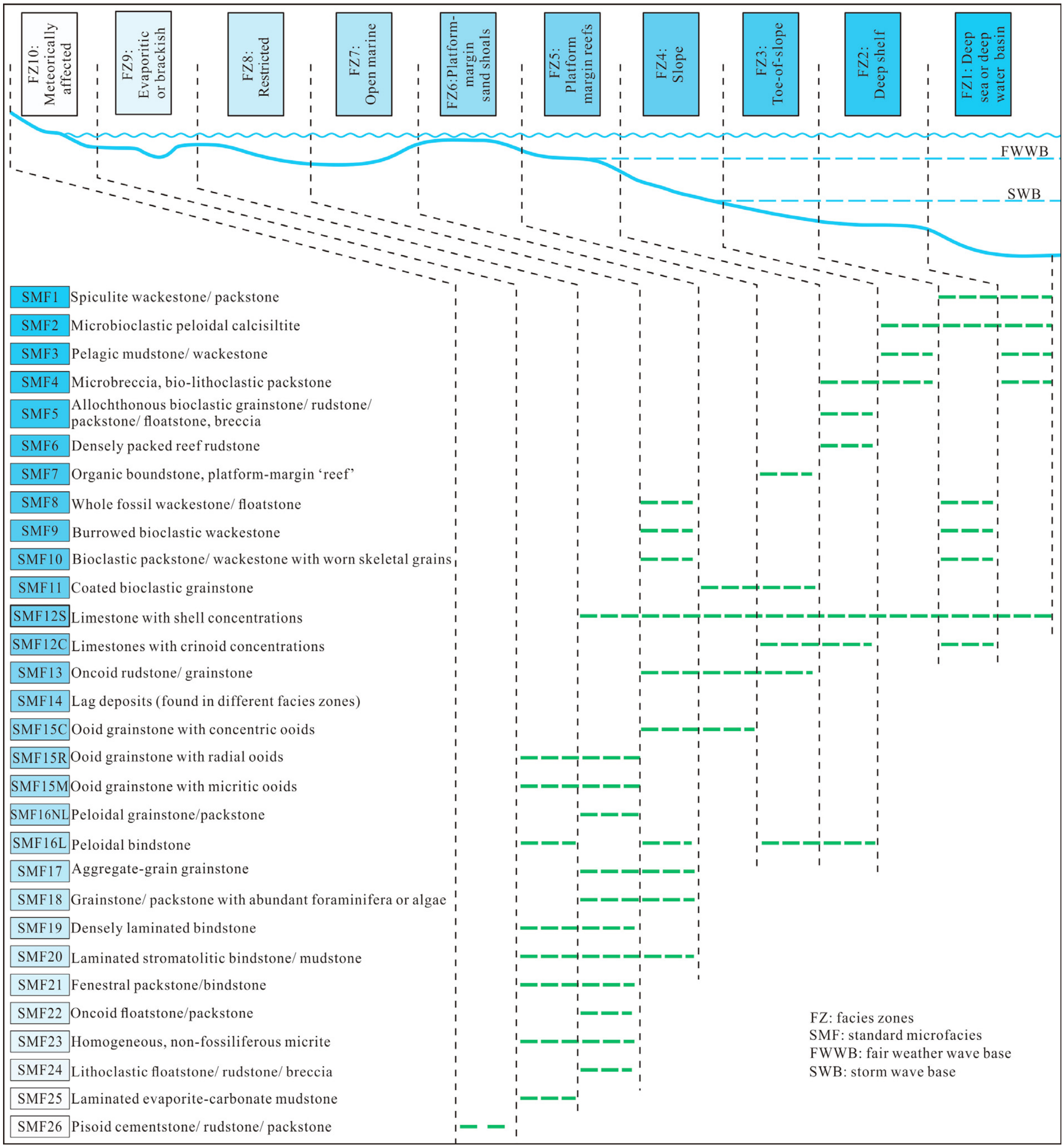

Fig. 1. Distribution of SMF in the FZ of the rimmed carbonate platform model (Flügel, 2004, 2010). Note: The darker the color of SMF and FZ boxes in the figure, the deeper the sedimentary environment.

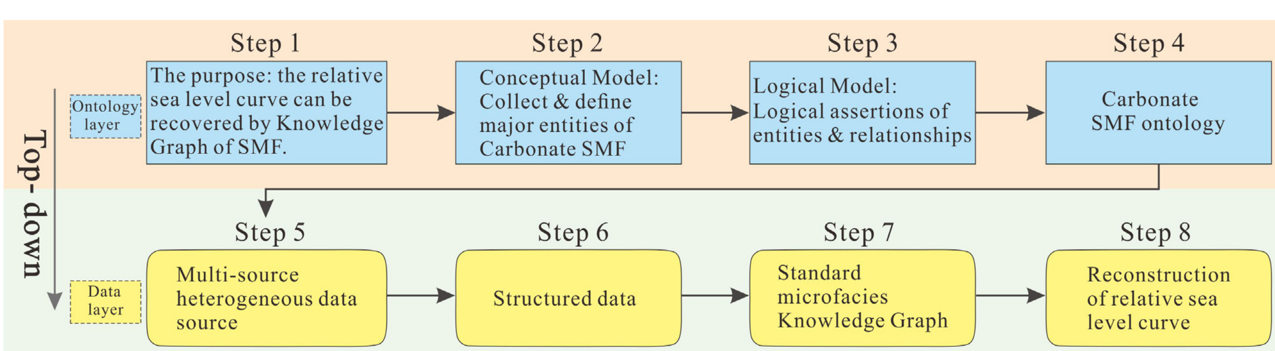

Fig. 2. The standard carbonate microfacies knowledge graph construction process. SMF: standard microfacies.

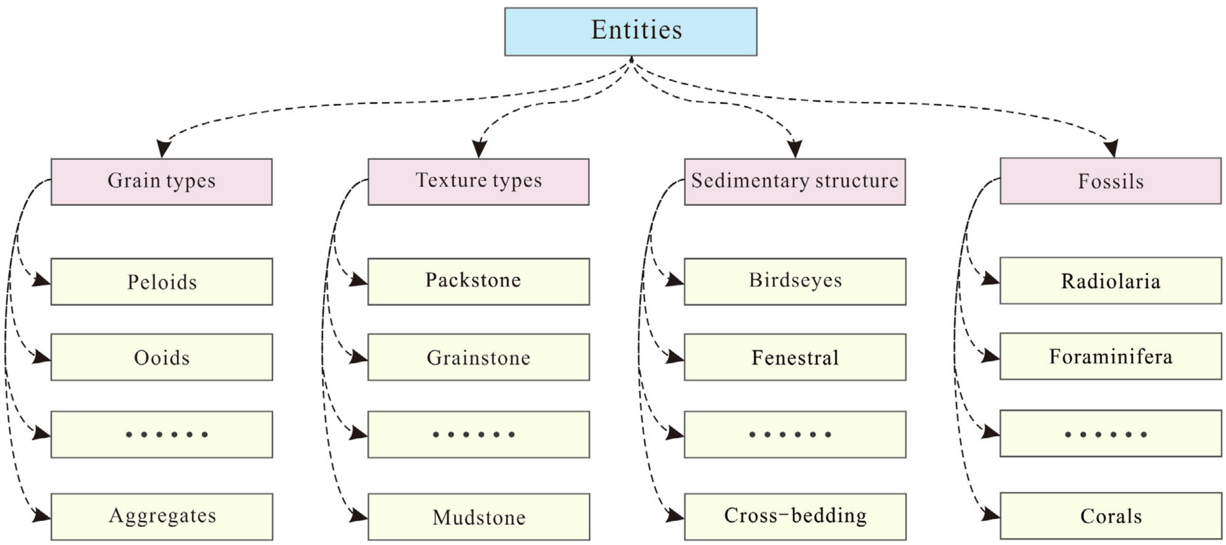

Fig. 3. Entities (partial) in standard carbonate microfacies ontology.

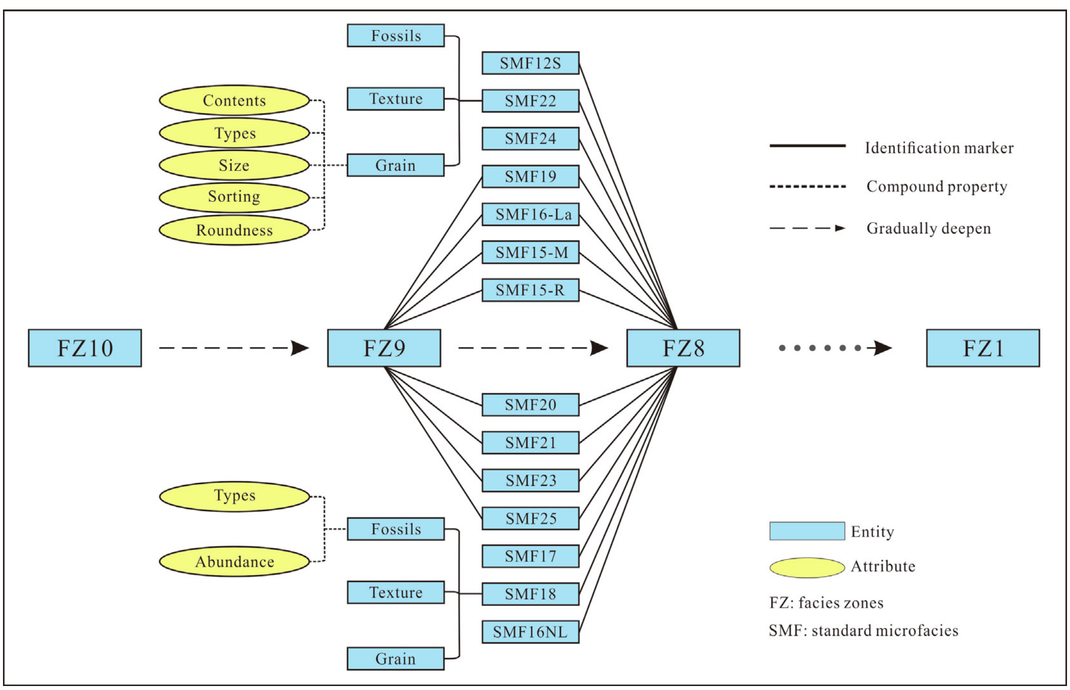

Fig. 4. Relationships (partial) among entities in the standard carbonate microfacies.

3.1.3. Collection of attributes

There are two kinds of attributes, as follows:

(1) Object attributes, such as grain sorting/rounding, biological species, matrix type, etc.; and
(2) Numerical attributes, such as grain size and content, biological abundance, and matrix content.

Additionally, each facies zone gradually deepens, which gives each facies zone a relative water-depth value (numerical attribute value). Similarly, the SMF also gradually deepen, which can also indicate the range of relative water depth. For example, from deep to shallow, the SMF1–SMF26 are assigned the relative water-depth values (SMFD26–SMFD1) and the FZ1–FZ10 are assigned the relative water-depth values (FZD10–FZD1), respectively.

3.2. Construction of the data layer

The data layer is a large amount of specific data corresponding to the ontology layer (Buccella et al., 2009, 2011; Chen et al., 2020). In the mass of knowledge, there are mainly structured, semi-structured and unstructured data (Kumar and Zayaraz, 2015; Hao et al., 2021). The data layer construction uses knowledge extraction technology to obtain effective information such as the entities, relationships, and attributes from the data sources and store it in the corresponding triples in the ontology layer (Knublauch et al., 2004; Bhattacharya and Getoor, 2007; Sahoo et al., 2009; Hao et al., 2021). There are two main sources for the data layer of the SMFKG. The first source is the text content of the description thin sections in carbonate microfacies analysis articles. Here, text extraction technology is used to obtain data (corresponding entity, relationship and attribute). Secondly, when observing the thin sections, the description contents are standardized and classified in advance according to the different concepts, relationships and attributes in the ontology design of the knowledge graph (Flügel, 2004, 2010). In this way, the main information corresponds to the ontology of the SMFKG designed above one by one, and the data layer is directly obtained and stored. This paper mainly adopts the second method. Table 1 shows the fragments of standardized CSV format data from the observation and description of thin sections.

3.3. Knowledge graph generation

Python is used to read the data and write to the Neo4j graph data. First, we created a new local database in Neo4j, named the database, and then imported the existing CSV format data into the py2neo library to generate a knowledge graph (Fig. 5). The SMFKG formed in this case has the basic application function of the knowledge graph. All database queries, SMF queries, or relationship queries, etc., can be carried out in Neo4j. For example, you can query the node of a specific SMF, from which you can intuitively query all of its IDM and all related FZ; or, query a certain IDM (such as oolite), and you can directly see which facies areas in which it is distributed.

4. Reconstruction of the relative sea-level curve based on the knowledge graph

4.1. The workflow and algorithm

By mapping the high-dimensional data to the low-dimensional space, the information about the knowledge graph will be expressed in the form of dense vectors, which provide a basis for clearly representing the knowledge graph (Hamilton et al., 2017; Nicholson and Greene, 2020). Once the knowledge graph is transformed into a low-dimensional vector space, machine-learning technology can use this space to construct query or reasoning applications (Grover and Leskovec, 2016; He et al., 2016). Therefore, in this study, the knowledge graph is first vectorized in low dimensions, and then the automatic identification of SMF and FZ application is constructed. Following this, the relative sea-level curve and high-frequency sea-level curve are automatically reconstructed according to the relative water depths represented by the different SMF and FZ, respectively.

4.1.1. Microfacies recognition

First, the knowledge graph is vectorized in low dimensions by word embedding, and the process of using IDM to infer SMF is constructed. Then, the structured or semistructured IDM obtained through thin section observations and descriptions are used for automatic identification of SMF and FZ.

4.1.1.1. Word embedding. Word embedding is the basis of the reasoning of a knowledge graph (Grover and Leskovec, 2016). In this work, word embedding is a process of vector representation of node words according to the connections among nodes of the knowledge graph in Neo4j (Kutuzov et al., 2018). Perozzi et al. (2014) proposed the method of generating node sequences by random walk and, along with natural language processing technology (word2vec), realized word embedding for social networks. The SMFKG in this study has similar structural characteristics to a social network. The training samples composed of the IDM–SMF– FZ sequences generated by random walk can be vectorized after being processed by the word2vec algorithm, thus completing the word embedding of the knowledge graph (Mikolov et al., 2013). The node vector after embedding has 256 dimensions, and after further dimension reduction by the *t*-sne method, the three-dimensional and two-dimensional characterization results of the SMFKG are obtained (Figs. 6 and 7; Perozzi et al., 2014). In three-dimensional space, IDM, SMF, FZ, SMFD and FZD nodes are obviously clustered in different spatial ranges (Fig. 6). Similarly, important nodes are marked in two-dimensional, showing that the SMF and FZ nodes (including their corresponding depth values SMFD and FZD) are mainly distributed in the upper left of the image and the various IDM of the SMF are mainly distributed in the lower right of the image (Fig. 7). The distribution clustering effect of various nodes is obvious, indicating that the three-dimensional and two-dimensional vector inherits the node relationship in the knowledge graph after word embedding (see Supplementary Data).

4.1.1.2. Automatic inference of node relationships. This work designs an automatic inference model of nodes and relationships in the knowledge graph by using a search algorithm (see Supplementary Data). The inference model imitates the hierarchical IDM– SMF– FZ structure in the SMFKG, calculates the distances between the node and all nodes in the next layer, and obtains the maximum probabilities of connectivity between the node and the nodes in the next layer. When using IDM to infer SMF based on the knowledge graph, it is typically necessary to combine multiple IDM to distinguish SMF. Considering that the IDM node clusters and the SMF node clusters are distributed at the lower right and upper left of the

Table 1
Standardized data (partial) in csv format.

Number	Texture	Grain types	Grain content	Grain size	Grain sorting	Grain rounding	Matrix types	Matrix content	Fossils types	Fossils content	Sedimentary structure
1	Wackestone	Bioclastic	Common	/	/	/	Micrite	Common	Crinoids brachiopods bryozoans	Common	/
2	Mudstone wackestone	Micro-bioclasts	Common	/	/	/	Micrite	Abundant	Siliceous sponge spicules	Abundant	Sponge oriented alignment
3	/	Peloids	A few	Fine	Well-sorted	Round	Micrite	Abundant	Ostracods foraminifera gastropods	A few	Laminated
4	Grainstones	Floatstones rudstones breccias	Abundant	Coarse	Bad	Angular	/	/	/	/	Sometimes cross-bedded
5	Grainstones	Oolite	Abundant	/	Well-sorted	Round	/	/	/	/	Cross bedding
6	Rudstones packstones	Pisoid	Abundant	>2.0 mm	/	/	/	/	/	/	Tepee structures

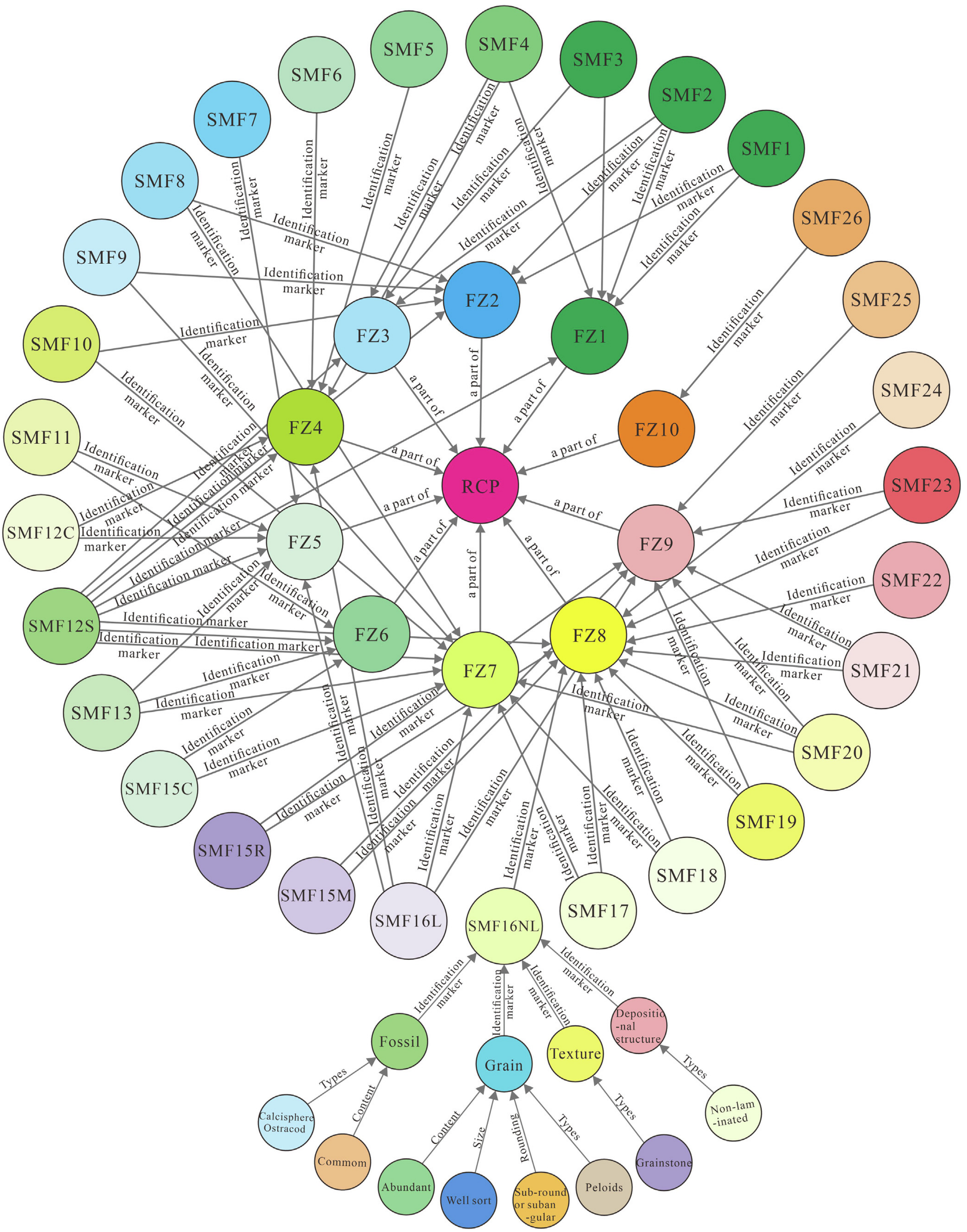

Fig. 5. The knowledge graph of standard carbonate microfacies (partial), in which RCP, FZ, and SMF represent rimmed carbonate platform, sedimentary facies zones, and standard carbonate microfacies, respectively.

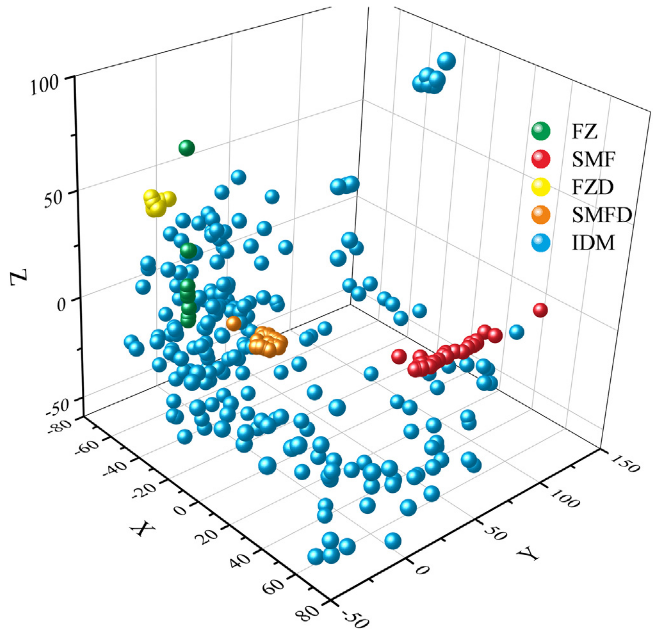

Fig. 6. Three-dimensional vector representation results of the standard carbonate microfacies knowledge graph (SMFKG). The green, red, yellow, orange, and blue circles represent the facies zones (FZ), standard carbonate microfacies (SMF), depth values of FZ (FZD), depth values of SMF (SMFD), and identification markers (IDM), respectively, in the SMFKG. It should be noted that the Fig. is the result of 50 times magnification and 85 degrees clockwise rotation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

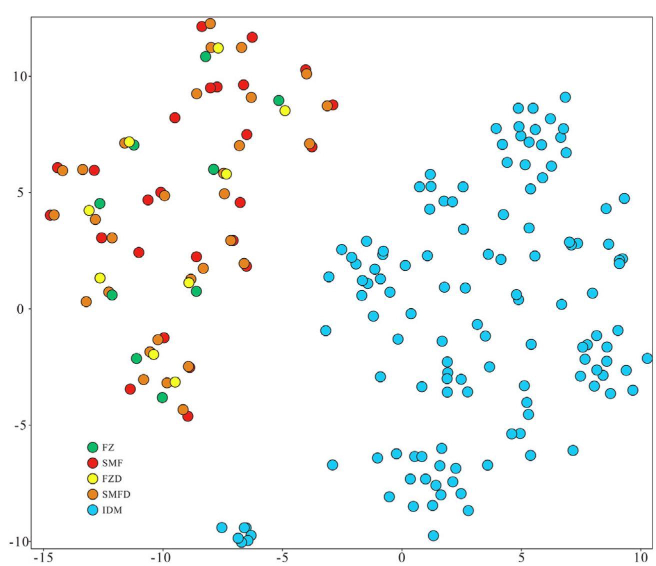

Fig. 7. Two-dimensional vector representation results of the standard carbonate microfacies knowledge graph (SMFKG). The color legends in the Fig. is the same as that in Fig. 6.

two-dimensional embedding results, respectively (Fig. 7), the IDM are projected first, and then the distances between their projections and the SMF nodes are calculated. The five specific steps are as follows: (1) calculate the sum S1 of the first two node vectors (F1, F2) in the IDM sequence; (2) calculate S1 and the next IDM node vector F3 to obtain S2 until the vector sum S_{n-1} of the last IDM node vector F_n in the sequence is obtained; (3) cyclically calculate the distances d (d_1, d_2,... d_i,... d_{26}) between S_{n-1} and the 26 SMF m (m_1, m_2,..., m_i,... m_{26}); (4) sort 1/d; and (5) process the sorting results using the softmax function, and the IDM and SMF connection results are obtained. Fig. 8 shows the comparative analysis between the automatic inference results of using the text of the thin-section descriptions and the actual SMF (obtained by manual analysis). In the diagonalized confusion matrix, the inference result obtained by vectorization of the IDM (*Y* direction) is almost consistent with the actual SMF (*X* direction), showing a diagonal distribution, which proves that the inference result has high accuracy (Fig. 8) (See for specific codes https://github.com/KeranLi/knowledge-graph/blob/main/Appendix.ipynb).

4.1.2. Distinguish sedimentary facies zones

In the rimmed carbonate platform sedimentary model, different FZ are composed of different SMF combinations, so the FZ can be determined according to the SMF combinations (Fig. 1; Flügel, 2004, 2010). There is one SMF only distributed in certain FZ or one SMF distributed in two, three, or more FZ at the same time so that a FZ may contain SMF with different occurrence frequencies (Fig. 9a). Assuming that a SMF is distributed in only one FZ, the probability value of distinguishing the FZ is 1. For SMF distributed in two FZ, the probability value of distinguishing the FZ is 0.8. For SMF distributed in three or more FZ, the probability value of distinguishing the FZ decreases further (Fig. 9a). Therefore, when a series of SMF is input and they all always exist in one FZ, the cumulative probability in this FZ is gradually increased and it is inferred that the probability of this group of SMF belonging to this FZ is the largest. When continuing to input the SMF, if the cumulative probability does not increase further, it is inferred that this SMF belongs to the next FZ (note that the FZ must be adjacent to each other and follow Walther's facies law) to judge the FZ cyclically.

The six specific steps are as follows: (1) input the SMF sequence k (k_1, k_2,..., k_n), and convert it into a word vector sequence; (2) calculate the distance d_i (d_1, d_2,... d_i,... d_{10}) between k_i and the word vector l (l_1, l_2,..., l_{10}) of all FZ nodes in the sequence; (3) set the distance amplification factor a = 2.5, sort $1/d_i \times 2.5$, and get the connection probability P_i (see the softxmax_dis function in the Supplementary Data for the calculation method); (4) record the FZ node corresponding to the maximum connection probability at this time, and create an array structure of k_i FZ and probability (see the f2smf function in Supplementary Data); (5) add k_{i+1} and k_i, calculate the connection probability for all FZ node word vectors, and generate a FZ/probability array P_{i+1} of k_{i+1}; and; (6) the

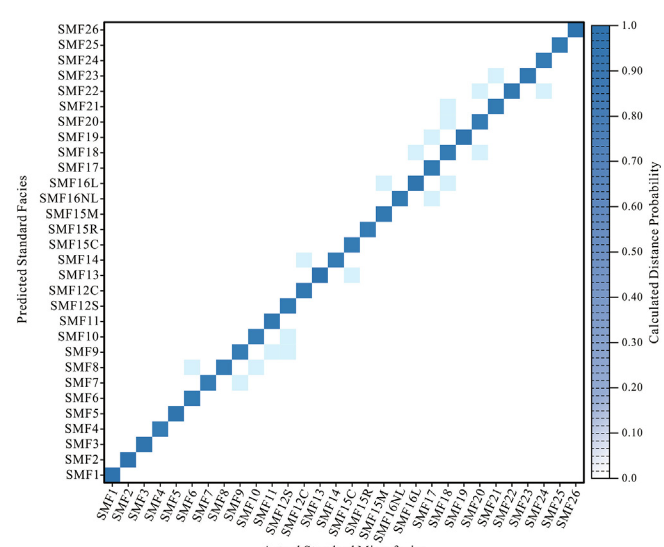

Fig. 8. Comparison of inferred microfacies by vectorization of identification markers with actual microfacies. SMF: standard microfacies.

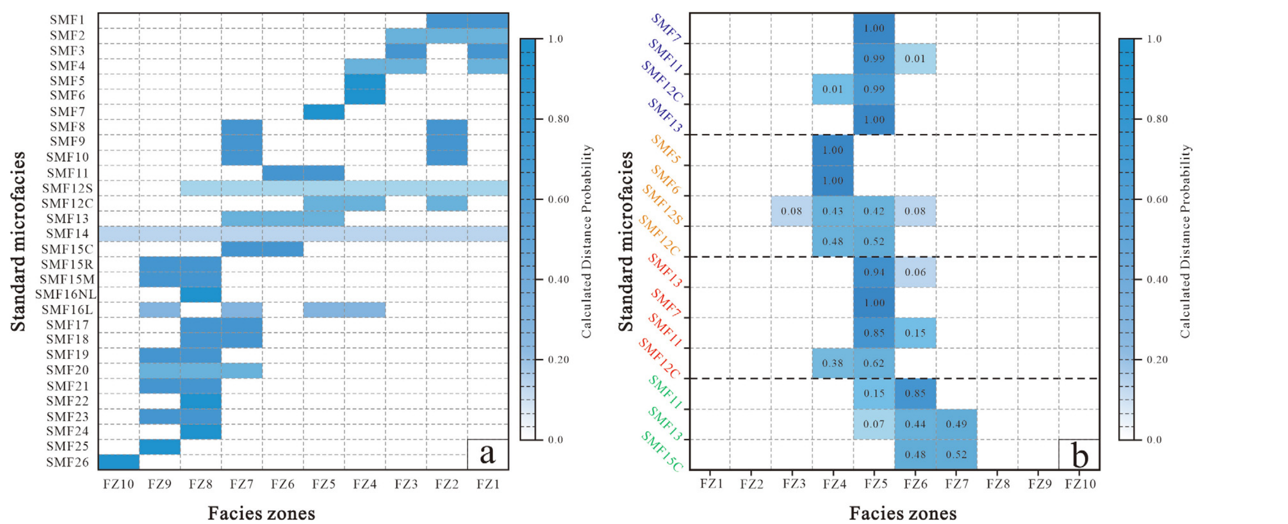

Fig. 9. (a) The distribution frequencies of SMF in different FZ in the rimmed carbonate platform. (b) Based on the law of SMF distribution in (a), 15 SMF (test data) are automatically divided into different FZ. SMF: standard microfacies; FZ: facies zone.

P_i and P_{i+1} rates are compared under the condition of the Walter's facies rule. When $P_i < P_{i+1}$, repeat steps (1)–(5); when $P_i \geq P_{i+1}$, output the FZ (See for specific codes https://github.com/KeranLi/knowledge-graph/blob/main/Appendix.ipynb).

In Fig. 9b, 15 SMF were tested and the input direction of the SMF is shown from bottom to top. When a single SMF is input, the corresponding FZ has multiple solutions. However, with the increase in SMF input, the probability of a unique FZ increases significantly. Finally, 15 SMF are automatically identified as four adjacent FZ (FZ6, FZ5, FZ4, FZ5 from bottom to top) (Fig. 9b).

4.1.3. Restoration of the relative sea-level curve

First, the sedimentary FZ in the rimmed carbonate platform have the characteristics of deepening from the coast to the deep sea. Therefore, given the relative water-depth value for each FZ, the changes in FZ can be used to roughly restore the relative sea-level variation curve. Additionally, the SMF in the rimmed carbonate platform has the behavior of gradually deepening or shallowing as a whole, so each SMF can represent a relative water-depth value. Therefore, based on the relative water depth represented by all SMF, the high-resolution relative sea-level variation curve can be directly reconstructed by the SMFKG. Additionally, the high-resolution relative sea-level variation curve recovered from the SMF can be compared and corrected with the relative sea-level variation curve recovered from the FZ.

The automatic reconstruction of relative sea level curve depends on the search algorithm of knowledge graph, searches the corresponding relationship between SMF (or FZ) and SMFD (or FZD) nodes in the knowledge graph and establishes the mapping relationship between them. The specific steps are as follows: (1) sort the SMFD nodes in the knowledge graph and establish the relative sea level depth sequence $d = \{d_1, d_2, \ldots, d_i\}$; (2) establish the predicted SMF node sequence $f = \{f_1, f_2, \ldots, f_i\}$; (3) based on the correspondence between SMF and SMFD nodes in the SMFGF, the conditional mapping of d sequence and f sequence is established; (4) based on the mapping relationship in (3), the visualization of sea level change curve is realized (The plotting code is accessible from https://github.com/KeranLi/knowledge-graph/blob/main/Appendix.ipynb). On this basis, the automatic reconstruction of the relative sea-level variation curve by using SMF and FZ is compiled (see the Supplementary Data for the detailed algorithm).

4.2. A case application

The sedimentary period of the Dengying Formation (approximately 551–541 Ma) in the Yangtze Platform is located at the Ediacaran–Cambrian (E–C) transition (Chen et al., 2015; Condon et al., 2005). Analyzing its relative sea-level variation curve is significant for the evolution of global paleostructure, paleogeography, paleoceanography, and paleontology at the end of the Precambrian. At the end of the Ediacaran, the Mianzhu Qingping area on the western margin of the Yangtze Platform is close to the edge of the proto-Tethys Ocean (Song et al., 2018; Qi et al., 2022), which is suitable for reconstructing the global relative sea-level curve. Additionally, the Dengying Formation is characterized by a stable rimmed carbonate platform, mainly composed of extremely thick (>800 m) dolomite deposits, and its relative sea-level change may represent global sea-level variation (Wang et al., 2020). Two regional subaerial-exposed unconformities exist in its interior and top, which can be accurately divided into two third-order sequences. These features make the Dengying Formation a good object for reconstructing the relative sea-level curve in deep time.

First, accurate stratigraphic measurements and a detailed sedimentological description of outcrops are carried out in the field, then, thin-section observations are carried out in the laboratory. According to the characteristics of lithology, color, composition, texture, and sedimentary structure, the Dengying Formation is divided into different sedimentary units from bottom to top, and fresh rock samples were collected to grind into thin sections. Each thin section was carefully observed, described, and analyzed using a polarized light microscope in the laboratory. Referring to Flügel (2004, 2010), all descriptions of each thin-section are classified according to the IDM of the SMF, such as grain type/content, matrix type/content, structure type, fossil type/abundance, or typical sedimentary structure, and accurately recorded to form a semistructured dataset. Using the SMFKG and its affiliated applications, the semistructured data of all thin sections are analyzed. From this, 17 SMF and 7 FZ are automatically identified.

Then, the relative sea-level variation curve at the end of the Ediacaran was roughly reconstructed using sedimentary FZ. The reconstruction results show that both the Lower and Upper Dengying Formation transition from FZ1 (deep-water basin) at the bottom to FZ7 (open platform)–FZ8 (limited platform) in the middle, to FZ10 (the subaerial-exposed unconformity) at the top. It can be seen from the FZ variations that the sedimentary period of the

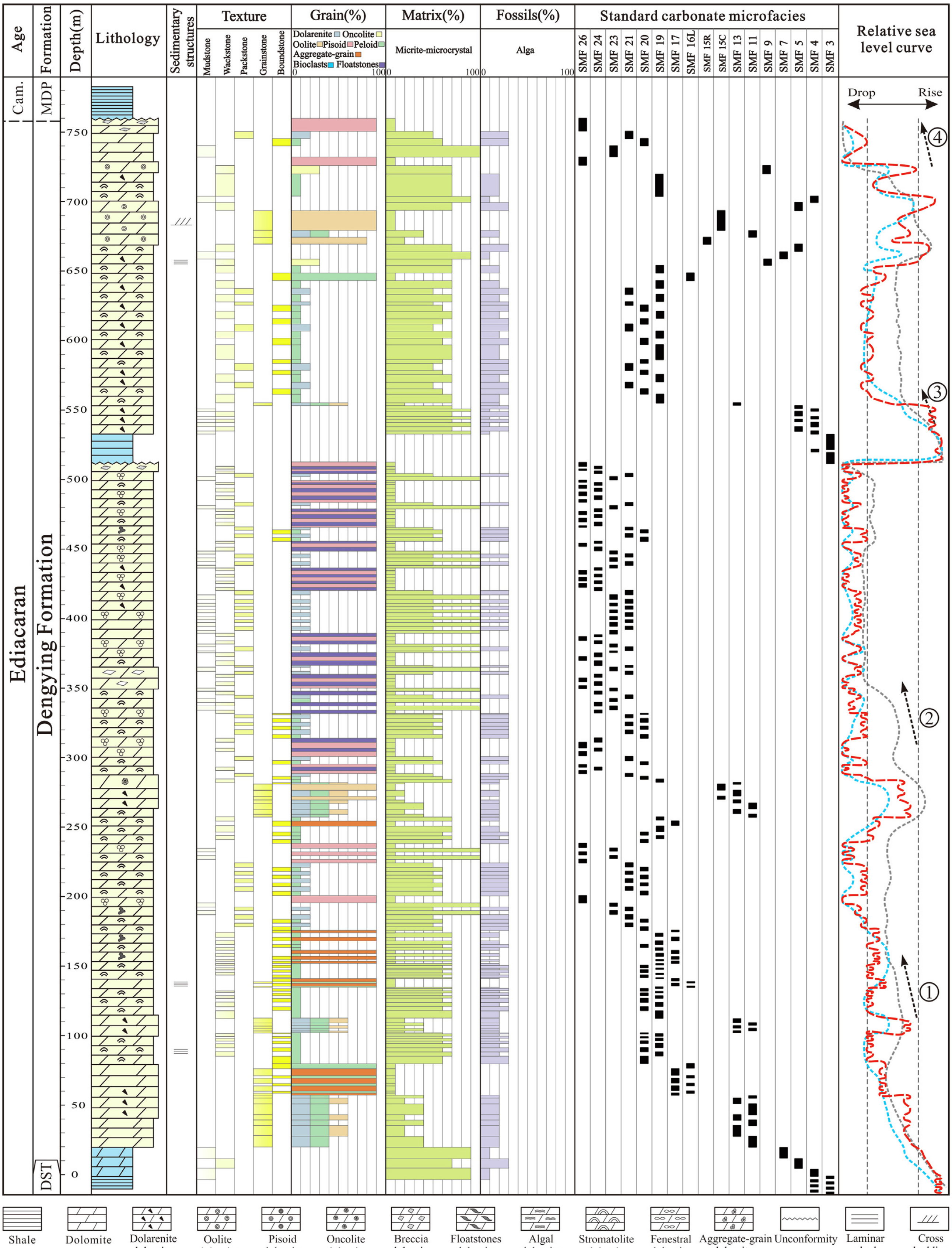

Fig. 10. Relative sea-level curve of the Dengying Formation at the end of the Ediacaran in the western margin of the Yangtze Platform reconstructed by the standard carbonate microfacies knowledge graph (SMFKG). The red and blue dashed lines on the right represent the high-resolution relative sea-level curve and secondary relative sea-level curve recovered from the SMF and FZ in the study, respectively; the gray dashed line represents the relative sea-level curve reconstructed by Qi et al. (2022), and the numbers in the circles represent the four sea-level changes. Cam: Cambrian; MDP: Maidiping Formation; DST: Doushantuo Formation; SMF: standard microfacies; FZ: facies zone. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Dengying Formation experienced two obvious sea-level rise and fall cycles (Fig. 10). Finally, according to the relative depth values represented by the SMF, the high-resolution relative sea-level variation curve at the end of the Ediacaran in the western margin of the Yangtze Platform is reconstructed. The sedimentary period of the Dengying Formation not only includes two obvious sea-level rise and fall cycles, but also has two secondary sea-level rise and fall cycle within them (Fig. 10). Therefore, four obvious sea-level rise and fall cycles occurred, among which two intense regressions led to the two subaerial-exposed unconformities in the interior and top of the late Ediacaran Dengying Formation (Fig. 10).

Qi et al. (2022) collected numerous thin-section samples from the same section of the Dengying Formation as in this study for carbonate microfacies analysis, then used microfacies to reconstruct the relative sea-level variation curve at the end of the Ediacaran. He identified 17 SMF and 4 kinds of self-defined microfacies through thin-section observation under a microscope (Qi et al., 2022). Among them, the 17 kinds of SMF are consistent with the results of automatic identification using the knowledge graph in this study, while the 4 kinds of self-defined microfacies that are inconsistent with this work may be caused by the differences of thin sections observation. Then, according to the relative water depth represented by the SMF, the relative sea-level curve in the late Ediacaran was manually restored, indicating four obvious sea-level rise and fall cycles (Qi et al., 2022), which is highly consistent with the automatic reconstruction results by using the SMFKG in this study (Fig. 10). Additionally, the algal dolomite of the Ediacaran Dengying Formation in the Yangtze Platform is a high-quality oil and gas reservoir, which has attracted in-depth research (Shi et al., 2018; Wei et al., 2018; Yang et al., 2018; Wang et al., 2020; Zhou et al., 2020; Yan et al., 2022). According to the latest oil and gas exploration data analyses (such as seismic, core, and well logging data), there are two subaerial-exposed unconformities within and at the top of the Dengying Formation, and then the Dengying Formation is divided into two third-order sequences, both of which contain two secondary sea-level variation cycles; thus, forming four sea-level rise and fall cycles (Jiang et al., 2011; Wen et al., 2016; Liu et al., 2017; Wang et al., 2019; Qi et al., 2022). Therefore, the reconstruction of the relative sea-level variation curve at the end of the Ediacaran using a knowledge graph is highly consistent with previous research results. Overall, this shows that automatic reconstruction of the high-resolution relative sea-level variation curve in deep time by using the SMFKG is highly efficient.

4.3. Prospect of the application

The automatic digital slide-scanning system (ADSS) has been widely used in the medical domain (Wong et al., 2018; Li et al., 2021). The latest ADSS (such as the Zeiss Axioscan 7) is an automatic instrument that integrates thin-section injection, loading, scanning, image processing, and image preservation (Yin et al., 2019). High-throughput automated optical petrography allows for thin sections to be fully digitized in batches of up to 100 simultaneously at unprecedented speed (Zingue et al., 2018). Integrating state-of-the-art machine-learning techniques with ADSS can automatically recognize the lithology, texture, sedimentary structures, fossils and diagenetic characteristics; make quantitative statistics (or measurements) of various minerals or paleontology; and convert all the thin sections into structured data (Filippis et al., 2020; Koeshidayatullah et al., 2020). These data can then be automatically imported into the powerful knowledge graph in this study (Filippis, 2019; Kazak et al., 2021), to automatically identify the SMF and FZ and automatically reconstruct the high-resolution relative sea-level variation curve in deep time. In other words, it is only necessary to provide numerous thin sections (from outcrops or boreholes) to fully automatically reconstruct the relative sea-level variation curve in deep time. Therefore, the combination of ADSS, machine-learning techniques, and the knowledge graph can realize the fully automatic microfacies recognition and reconstruction of high-resolution relative sea-level variation curves in deep time, which has considerable application value.

5. Conclusion

This work proposes a knowledge graph to identify SMF and reconstruct the high-resolution relative sea-level variation curve in deep time efficiently and intelligently. A comprehensive analysis is performed of the concepts, relationships, and attributes of SMF in various literature, and the ontology layer is constructed. The data layer is constructed using the data and text content obtained by describing the thin sections. Then, the SMFKG is constructed from top to bottom. Based on the SMFKG, the workflows and applications for automatically identifying SMF using the thin-section descriptions and for reconstructing the relative sea-level variation curve using the SMF and FZ are compiled. The case application results show that the Yangtze Platform underwent four sea-level rise and fall cycles in the late Ediacaran, which is highly consistent with previous research results, indicating that using the knowledge graph is an efficient and intelligent method to reconstruct sea-level curves in deep time. It should be specifically noted that the automatic reconstruction of the relative sea-level curve using the SMFKG is carried out under ideal conditions in this work and may need further improvement in practical applications. However, soon, the combination of ADSS, machine learning techniques and the knowledge graph in this study could provide an efficient method for microfacies recognition and the reconstruction of high-resolution relative sea level variation curves in deep time, which has broad application prospects.

CRediT authorship contribution statement

Han Wang: Writing – original draft, Writing – review & editing. **Hanting Zhong:** Writing – review & editing. **Anqing Chen:** Conceptualization, Supervision, Formal analysis. **Keran Li:** Software, Data curation, Visualization. **Hang He:** Software, Validation. **Zhe Qi:** Data curation. **Dongyu Zheng:** Software, Visualization. **Hongyi Zhao:** Software, Formal analysis. **Mingcai Hou:** Project administration, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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